



“Engineering students are choosing the easier path to employability compared to semiconductor-hardware engineering, which requires a master’s degree and years of experience”

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“There will be strong demand on recruiting processing engineers, equipment engineers, IC testing engineers and capacity planning managers”

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lack significant industrial fab experience. “The challenge now is to spread awareness and develop the faculty in this domain. The industry needs to pitch in, to curate the syllabi, and for the development of the faculty via its intrinsic expertise. This is a work in progress for the government, industry and the academia alike,” says Anurag Awasthi, VP at India Electronics and Semiconductor Association (IESA).

Although academic institutions are coming up with courses focussed on the sector, a primary challenge for them will be in providing hands-on and industry-relevant training in live environments. “The most important aspect of these diploma courses for manufacturing technicians is to get practical training, which should be a part of the course work itself,” says VLSI Society’s Gupta. The solution, he suggests, is that government and academic institutes should work with the industry to provide 8-12 weeks of internship at a running facility. “We can utilise the facilities at SCL (Semiconductor Lab) in Mohali or create tie-ups with fabs in Taiwan, Singapore

and Malaysia for real hands-on training,” he adds.

Even Hitesh Garg, India Country Manager at chip manufacturer NXP Semiconductors, says, “To boost employment, it is essential that engineering institutes focus on facilitating programmes that concentrate primarily on semiconductor equipment operation and manufacturing. These programmes can offer practical experience and hands-on training that will help them to explore jobs in various fields like process engineers, foundry engineers, etc.”

Acknowledging the need for skilled labour, many global institutes and organisations such as SEMI, Micron, Purdue, etc., are starting to offer courses and collaborations in this field. Not only that, institutes in India, such as IIT Mumbai, IIT Roorkee, IIT Madras, Sastra Deemed University, etc., are also partnering with global institutes, fabs and equipment companies to produce the right talent at various levels.

LEARNING ON THE JOB

“Given the variety of job roles, there is no single course. Just like in any other field, the B Tech and master’s-level courses offer a base, but much of the fab jobs also need on-the-job-training,” says independent semiconductor analyst Arun Mamphazy.

While a PhD or master’s is required for some high-end jobs, chip engineers mostly join fabs post-completion of their courses, and then up-skill on the job. Also, as the demand for highly skilled talent regularly outstrips its supply, the semiconductor industry invests significantly to maintain and advance its workforce’s skills.

For example, a major US semiconductor firm runs a robust \$1.4-million annual educational assistance programme through which hundreds of employees are supported each year to pursue master’s, bachelor’s, associate-, and certificate-level programmes in their job-related fields. Similarly, another firm supports its employees’ up-skilling needs with a \$30,000 annual tuition reimbursement programme. Beyond tuition assistance, one US firm hires over 170 student interns each year, while another firm spends over \$1.5 million annually to support approximately 70 interns across its operations in the US.

Indian conglomerate Vedanta, which is in the process of setting up a chips fab (in partnership with Taiwan’s Foxconn) and a display fab in the country, has also chalked out a plan to hire experienced personnel for its fab. As Vedanta will be setting up a greenfield project, the company plans to bring the world’s best talent initially, and then look to develop talent in India.